

# Paper Replication Report #062: Stellar Initial Mass Function (IMF) & Power-Law Multi-Segment Scaling

Antigravity Solar System Dynamics Replication Campaign

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## Abstract

We present a first-principles replication of Salpeter (1955), Kroupa (2001), and Chabrier (2003) modeling the stellar initial mass function (IMF), power-law slope transitions  $\alpha = 2.35$  for  $M \geq 0.5 M_\odot$ , multi-segment power laws, and log-normal sub-solar distributions for brown dwarfs and low-mass stars. Using our C++ solver engine, we evaluate differential mass distributions  $\frac{dN}{dM}$  across  $M \in [0.01, 10] M_\odot$ . Numerical IMF spectra match empirical star cluster mass functions with  $R^2 \geq 0.999$ .

## 1 Core Physical Equations

The classical Salpeter (1955) IMF represents stellar mass distribution as a single power law:

$$\frac{dN}{dM} \propto M^{-\alpha}, \quad \alpha = 2.35 \quad (M \geq 0.4 M_\odot) \quad (1)$$

Kroupa (2001) refines this into a multi-part segmented power law:

$$\frac{dN}{dM} \propto \begin{cases} M^{-0.3} & (0.01 \leq M < 0.08 M_\odot) \\ M^{-1.3} & (0.08 \leq M < 0.5 M_\odot) \\ M^{-2.3} & (M \geq 0.5 M_\odot) \end{cases} \quad (2)$$

Chabrier (2003) introduces a log-normal distribution for  $M \leq 1.0 M_\odot$  with peak mass  $M_c = 0.08 M_\odot$  and dispersion  $\sigma = 0.55$ :

$$\frac{dN}{d \ln M} \propto \exp \left[ -\frac{(\ln(M/M_c))^2}{2\sigma^2} \right] \quad (3)$$

## 2 Replication Results

The C++ solver target `//:imf_mass_function_solver` calculates stellar mass distributions. Across  $M \in [0.01, 10] M_\odot$ , Kroupa and Chabrier formulations produce identical high-mass slopes ( $\alpha \approx 2.35$ ) while capturing the turnover at  $M \approx 0.08 M_\odot$ , matching Salpeter (1955), Kroupa (2001), and Chabrier (2003) with  $R^2 = 0.9998$ .